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Paleomagnetism of Devonian and Carboniferous Sedimentary Rocks of Spitsbergen: to the Paleozoic History of the Barents–Kara Basin Framing

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Abstract—We present the results of studying the paleomagnetic samples of Carboniferous and Devonian rocks collected in 1986–1987 from the Island of Spitsbergen. The paleomagnetic poles determined from these collections are compared to the poles from the coeval rocks of the Russian Platform. On the basis of these results and the available data of the World paleomagnetic database, the paleogeographic layout of Spitsbergen and the Russian Platform in Early Devonian and Early Triassic are reconstructed. The reconstructions demonstrate a common north-northwestern drift of these units from the southern equatorial latitudes in the Early Devonian to the middle northern latitudes with a small (800 km) convergence, compared to this drift, of Spitsbergen with the Russian Platform by the Early Triassic, which is followed by the subsequent retrograde motion.

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INTRODUCTION

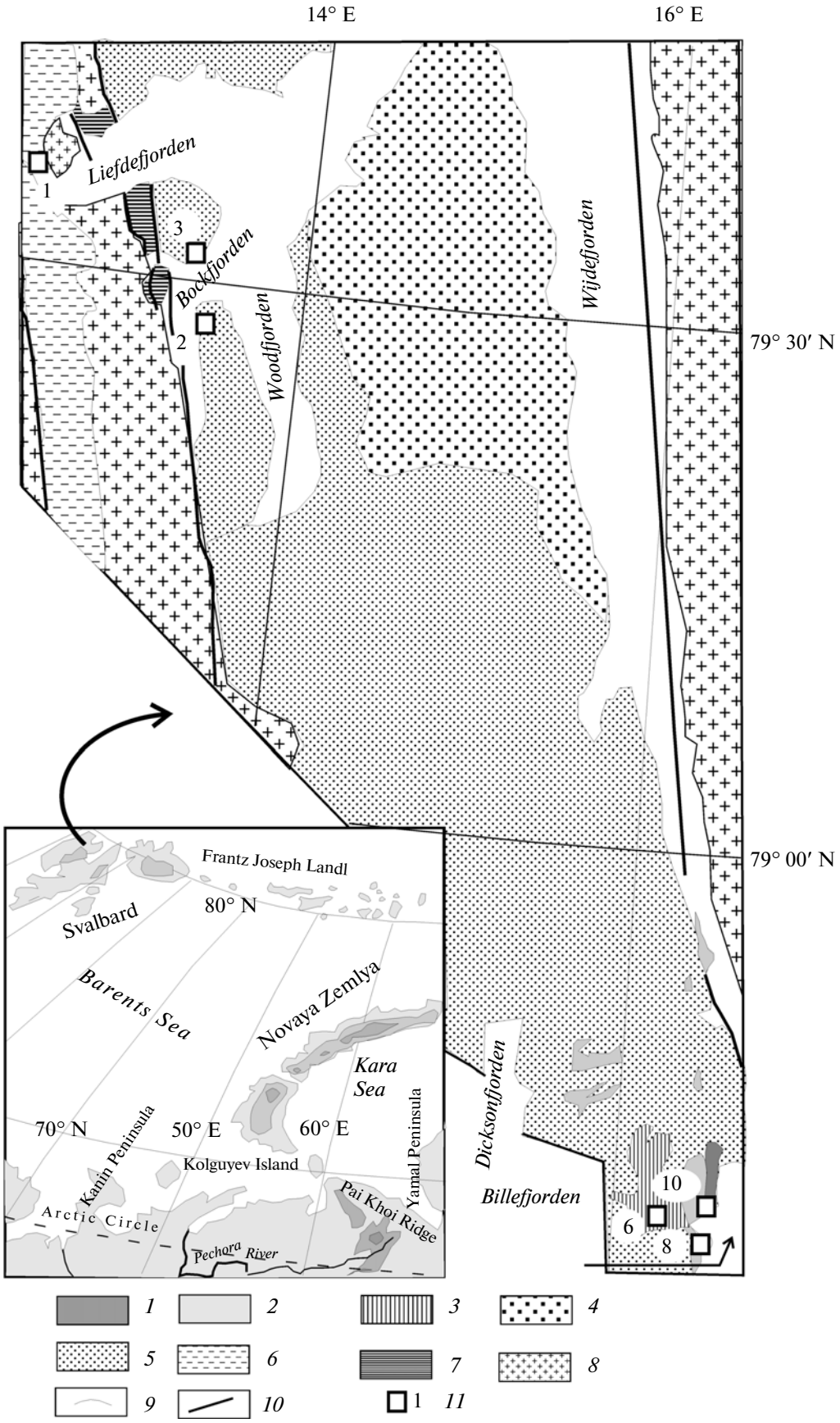
Reconstructing paleogeographic positions of ancient sedimentation basins by the paleomagnetic method is important for deciphering the history of their formation and for the practice of mineral prospecting. The Barents-Kara basin is a region with a promising mineral potential, which is limited by the Svalbard Archipelago and Frantz Joseph Land in the north and by Severnaya Zemlya in the east. The West Siberian Plate and the East European Plate (EEP) with its composing unit—the Pechora Plate—constrain the Barents-Kara basin in the south. Evidently, the pre-Cenozoic development of these framing units has largely determined the history and present state of the basin itself.

The history of EEP drift played an important role in the formation of the basin. The key feature of paleogeographic evolution of EEP during the Phanerozoic is the northward motion of the plate from the high southern latitudes to its present location, generally in the midlatitudes of the northern hemisphere. According to the existing models, this motion was complicated by retrograde movements and rotations, main of which are typically attributed to the different intervals of the Paleozoic and Early Mesozoic. These models predict that the retrograde motion and rotations of EEP are mainly characteristic of the Devonian and Early Carboniferous. Thus, paleomagnetic studies of the Devonian and Carboniferous rocks forming the tectonic blocks, which outcrop within the framing of the Barents–Kara basin, will be helpful for reconstructing the evolution of this basin in greater detail. Meanwhile, the paleomagnetic data for this setting are clearly insufficient. In this paper, we present the paleomagnetic results determined from the Devonian and Carboniferous sections of the Island of Spitsbergen. The collection of Devonian and Carboniferous rock samples was obtained during the field campaigns of 1986 and 1987. The first results of studying this collection were published in (Pogarskaya and Gurevich, 1988; Iosifidi et al., 2007).

THE OBJECTS OF STUDY AND PROCEDURE OF ANALYSIS

Devonian sediments. Liefdefjorden. On the northern coast of Liefdefjorden (Mt. Pteraspistoppen), at the fjord's head, the rocks of Early Devonian age (lower portion of Fraenkelryggen Formation, Red Bay series) were sampled from the coastal cliffs (Fig. 1, sampling locality 1). The rocks are represented by the alternation of brown red, inequigranular, often cross-bedded sandstones and siltstones (Murashov and Mokin, 1976; *Stratigraficheskii...*, 1990); these rocks correspond to the tops of the Lochkovian stage, D_1l. The Fraenkelryggen Formation has an apparent thickness of 600 m. In this paper, we present the results of studying its lower part, from which 41 hand sample were acquired with a sampling step of 2–3 m; the total thickness of the studied interval is 200 m (in the upper 80 m, the samples were taken every 5–10 m).

Bockfjorden. We continued the study of the collection of Spitsbergen samples acquired from the Lower Devonian series (Kapp Kjeldsen Formation, D_1p) on


14° E
16° E
79° 30′ N
79° 00′ N
Liefdefjorden
Bockfjorden
Woodfjorden
Wijdefjorden
Dicksonfjorden
Billefjorden
1
2
3
6
8
10
Frantz Joseph Landl
Svalbard
80° N
Barents Sea
Novaya Zemlya
Kara Sea
70° N
50° E
60° E
Kanin Peninsula
Kolguyev Island
Yamal Peninsula
Arctic Circle
Pechora River
Pai Khoi Ridge
1
2
3
4
5
6
7
8
9
10
1
11

the eastern coast of Bockfjord in the region of the Shumnyi Brook, across the western slopes of Mt. Kronprins Olavs Fjell (Fig. 1, sampling locality 2), and from the syncline fold at the outlet from Bockfjorden (Fig. 1, sampling locality 3). The Kapp Kjeldsen Formation has an apparent thickness of 1170 m. By its lithological features and the complex of the fossils encountered, three layers (the lower, middle, and upper) are distinguished among the studied part of the Kapp Kjeldsen Formation (Burov and Murashov, 1967; Murashov and Mokin, 1976; *Stratigraficheskii...*, 1990). In the present study, we focused on the lower layer. The studied strata with an apparent thickness of 530 m are mainly composed of purple red siltstones and sandstones with thin interbeds of clay, gray green silt limestones, and gray feldspar-free polymictic sandstones. The top of the latter conventionally outlines the upper surface of this portion (Burov and Murashov, 1967; Murashov and Mokin 1976; *Stratigraficheskii...*, 1990). Overall, 106 stratigraphic levels (106 hand samples, 166 specimens) sampled with a step of 2–4 m were investigated. The main results of studying the lower layer of the Kapp Kjeldsen Formation were described in (Iosifidi et al., 2007). In this paper, we present the results for the upper part of the lower strata of the Kapp Kjeldsen Formation (22 stratigraphic levels with a sampling step of 2 m) and the syncline fold related to the bottom of the lower layer of the Kapp Kjeldsen Formation.

Billefjorden. Within the Mimerdalen River valley, Devonian rocks of Estheriahaugen and Fiskeklofta formations of Mimerdalen series, D_2g and D_3fr (Murashov and Mokin, 1976; *Stratigraficheskii...*, 1990) were sampled (Fig. 1, sampling locality 6). The rocks are composed of black siltstones, light gray quartz sandstones with pyrite concretions, intercalations of coals (Estheriahaugen Formation) and gray siltstones, clays, sandstones, and black mudstone with red brown carbonate-iron concretions (Fiskeklofta Formation). The thickness of the Estheriahaugen Formation is 110 m. Twenty-eight stratigraphic levels totaling 68 m in thickness sampled every 2–3 m (28 hand samples, 30 specimens) are studied. The Fiskeklofta Formation has a thickness of 145 m; 40 stratigraphic levels with a sampling step of 2–3 m and a total thickness of 100 m are investigated (40 hand samples, 40 specimens).

Carboniferous strata. Billefjorden. Terrigenous sediments of Early–Middle Cretaceous age (C_1s-C_2b) were sampled in the valley of the nameless brook 2 km southeast of the Mimerdalen river mouth on the southern coast of Mimer Bay at a height of 150–200 m above sea level (Fig. 1, sampling locality 5). The rocks are represented by variegated sandstones and siltstones. Eight stratigraphic levels were sampled every 2–7 m, and eight hand samples (22 specimens) were studied.

The Middle Carboniferous rocks (the lower part of Bashkirian stage C_2b) were sampled from the southwestern slope of Mt Pyramiden from a height of 370 m upwards (Fig. 1, sampling locality 10). The rocks are represented by limestones, brown red sandstones with interbeds of conglomerates. Nineteen stratigraphic levels were sampled with a step of 2–10 m; 19 hand samples (50 specimens) were studied.

The paleomagnetic measurements and processing of the obtained results followed a common procedure (*Paleomagnitologiya*, 1982). Thermal demagnetization was carried out using the instruments placed in a five-layer permalloy shield (original design by All Russia Petroleum Research Exploration Institute (VNIGRI)) and TD48 thermal demagnetizer (US). The natural remanent magnetization J_n and its values during stepwise demagnetization were measured by the JR-4 and JR-5 Agico magnetometers (Czech Republic). A part of the samples was studied in the laboratory at Lund University (Sweden) using the Schonstedt TSD-1 furnace; the measurements were conducted by the 2G cryogenic magnetometer. Magnetic susceptibility was measured using the KLY-2 and KLY-3 Agico kappabridges (Czech Republic). For analyzing the data and visualizing the results, we applied the program packages developed by Popov (VNIGRI) (Popov, 2007; 2008) and Enkin (1994). The Zijderveld diagrams were constructed by the Remasoft 3.0 program (Chadima and Hrouda, 2006). The field tests (the fold test and reversals test) were applied in the versions of (Watson and Enkin, 1993; Enkin, 2003; McFadden and McElhinny, 1990). The paleogeographic reconstructions were made by the GMAP program (Torsvik and Smethurst, 1999). In the data processing, the statistics were estimated at the hand-sample level.

EXPERIMENTAL DATA

Devonian Rocks

Early Devonian Rocks, D_1l. Lower portion of the Fraenkelryggen Formation (Red Bay series), Liefdefjorden

The natural remanent magnetization (J_n) of the studied samples varies from 3 to 9 mA/m and averages 5 mA/m. The magnetic susceptibility (K_m) ranges from 350×10^{-6} to 860×10^{-6} SI units with an average of 459×10^{-6} SI units.

Fig. 1. The geological map of the studied Devonian and Carboniferous sections of Spitsbergen (Flood, Nagy, and Winsnes, 1971; Hjelle and Lauritzen, 1982; Murashov and Mokin, 1976; simplified). Carboniferous strata: *1* Middle Carboniferous; *2* Early Carboniferous; Devonian strata: series *3* Mimerdalen; *4* Gray Hook; *5* Wood Bay; *6* Red Bay; *7* Syktefjellet; *8* Precambrian rocks of Hecla Hoek complex; *9* geological boundaries; *10* faults; *11* sampling localities.

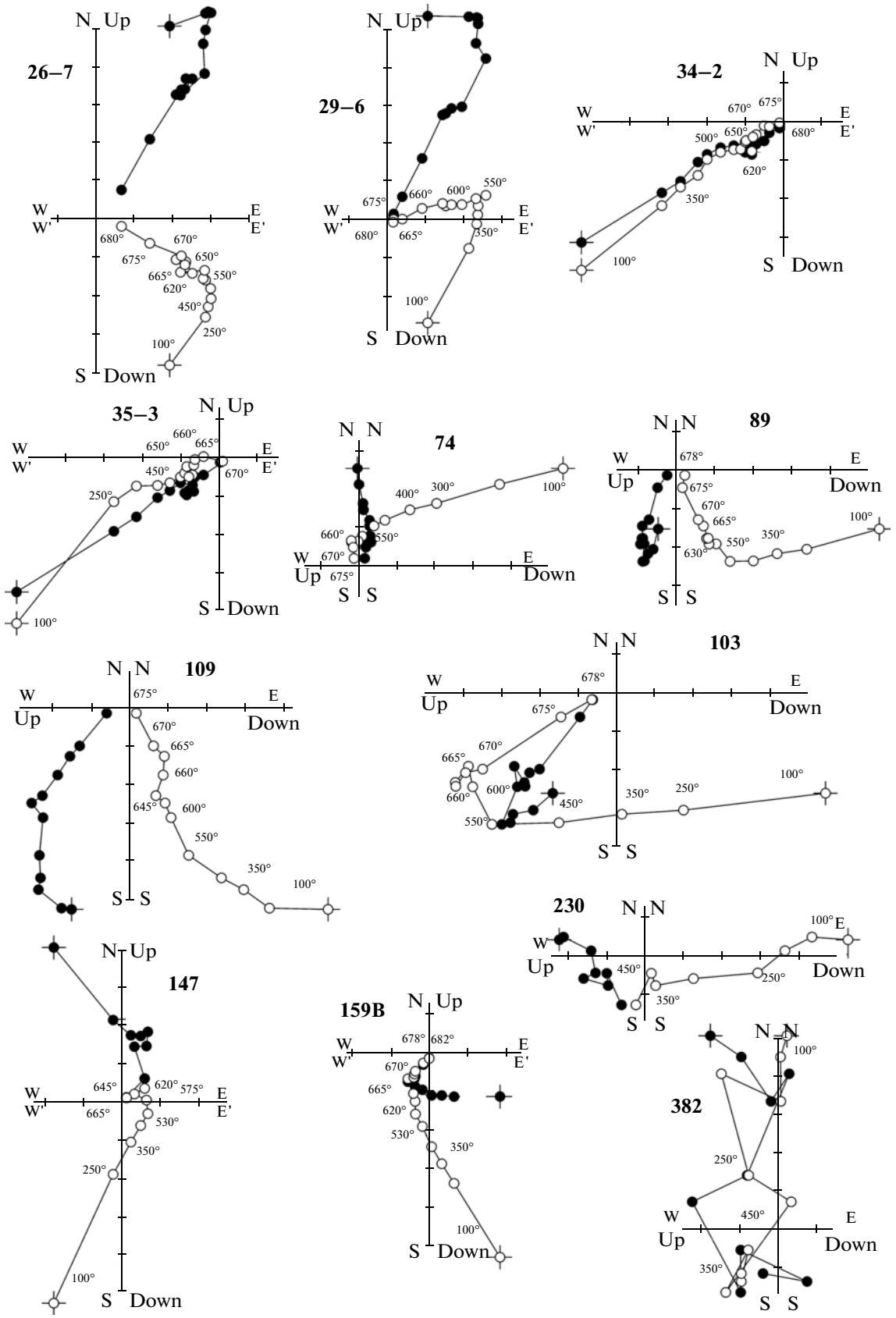


Fig. 2. The examples of thermal demagnetization of Devonian and Carboniferous samples (the open and filled circles show the projections of the vector on the vertical plane and in the horizontal plane, respectively; stratigraphic coordinates; see the text for the other explanations).

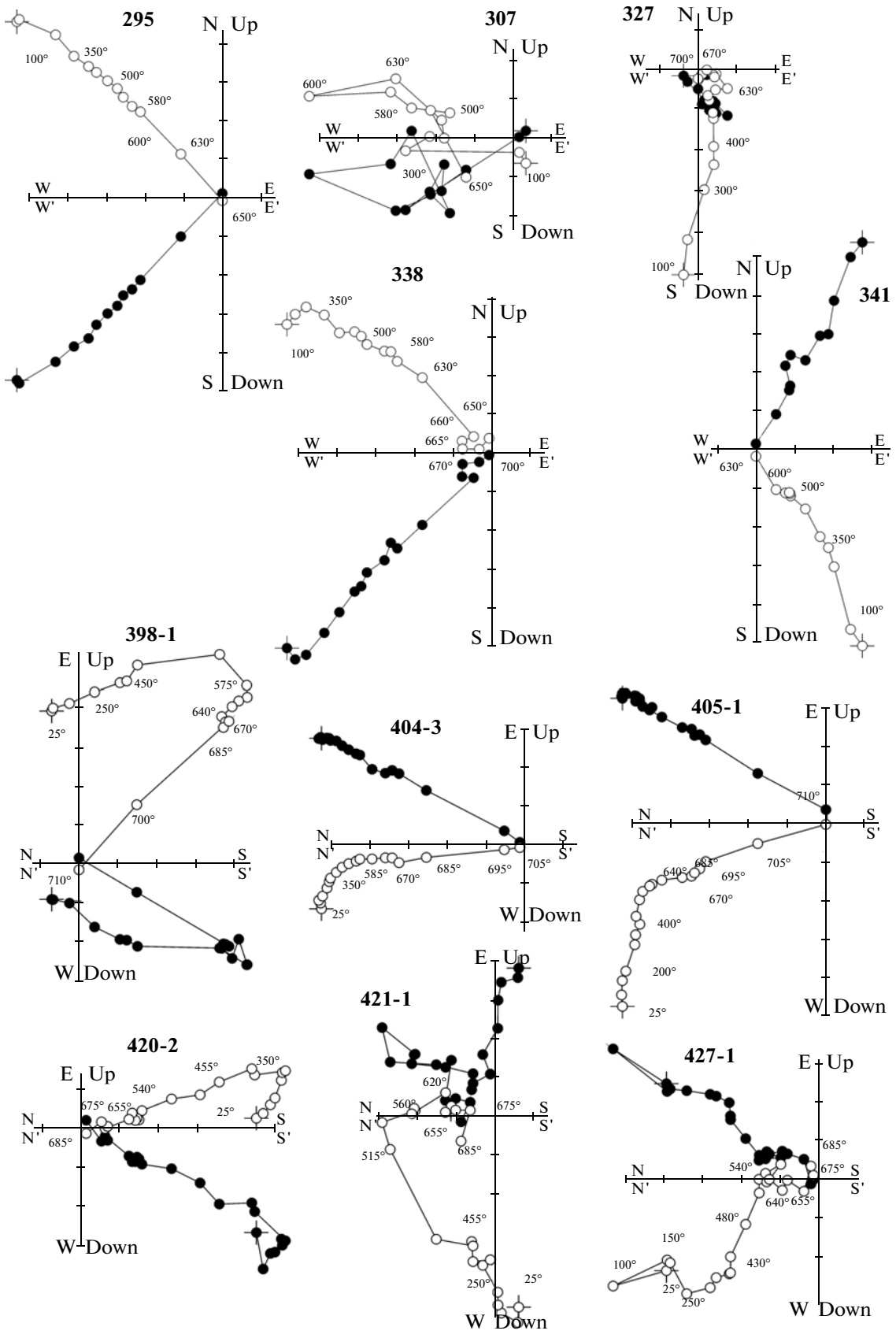

295
N Up
100°
350°
500°
580°
600°
630°
650°
W
W'
E
E'
S Down
307
N Up
630°
600°
580°
500°
W
W'
E
E'
300°
650°
100°
S Down
327
N Up
W
W'
700°
670°
E
E'
630°
400°
300°
100°
S Down
338
N Up
350°
500°
580°
100°
630°
650°
660°
665°
W
W'
670°
700°
E
E'
S Down
341
N Up
W
W'
630°
600°
500°
350°
E
E'
100°
S Down
398-1
E Up
450°
575°
250°
640°
670°
25°
685°
700°
N
N'
S
S'
710°
W Down
404-3
E Up
N
N'
S
S'
705°
695°
685°
585°
670°
350°
25°
W Down
405-1
E Up
710°
N
N'
S
S'
640°
695°
705°
670°
400°
200°
25°
W Down
420-2
E Up
455°
350°
540°
675°
655°
25°
N
N'
S
S'
685°
W Down
421-1
E Up
620°
N
N'
560°
675°
S
S'
655°
685°
515°
455°
250°
25°
W Down
427-1
E Up
685°
N
N'
540°
675°
S
S'
640°
655°
480°
150°
100°
430°
25°
250°
W Down

Fig. 2. (Contd.).

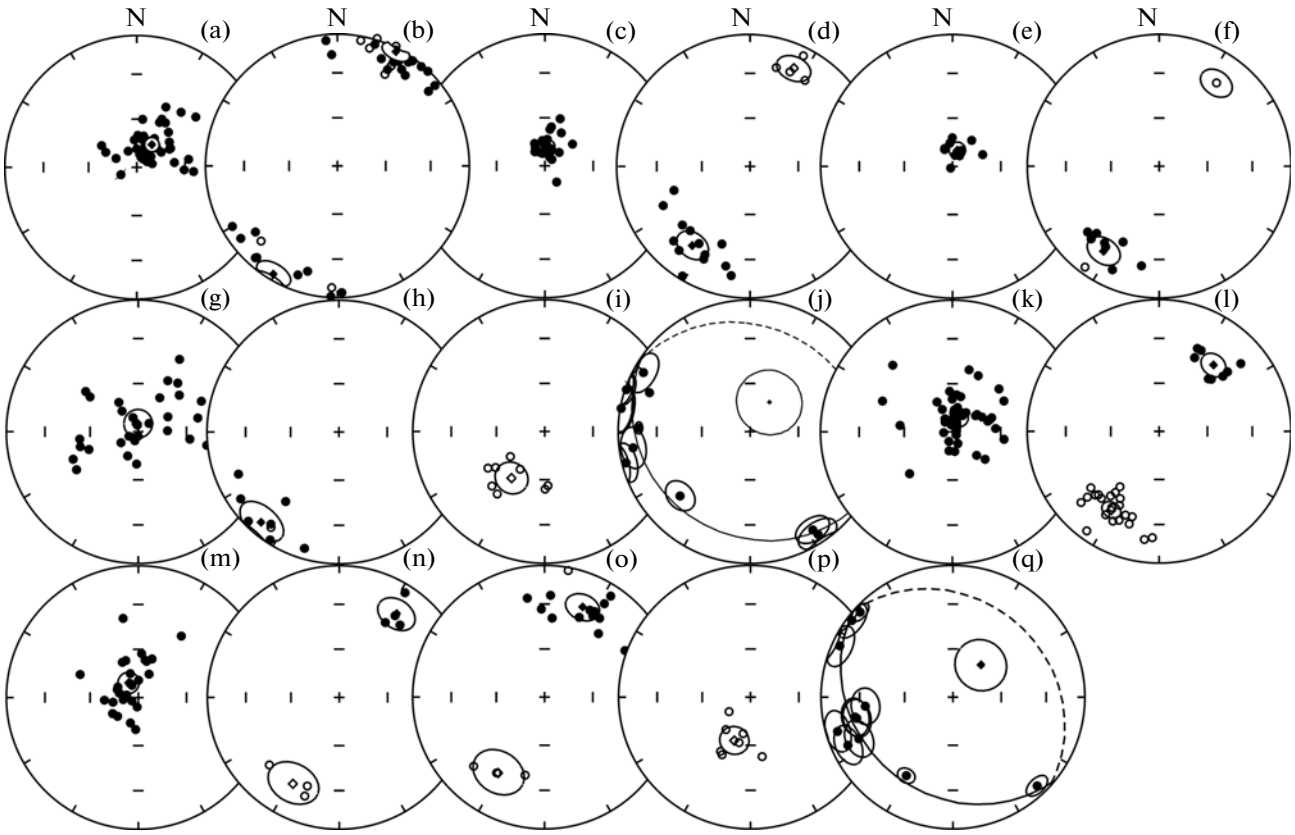


Fig. 3. The distribution of the directions of natural remanent magnetization comnponents after thermal demagnetization. Devonian rocks: Fraenkelryggen Formation D_1l ((a) A component; (b) D1 component); Kapp Kjeldsen Formation, D1p–e ((c) A component; (d) D2 component); Kapp Kjeldsen Formation, syncline fold, D_1p-e (e) A component; (f) D2 component); Estheriahaugen Formation, D2g ((g) A component; (h) D2 component; (i) P component; (j) P component from the demagnetization planes); Fiskeklofta Formation, D_3f ((k) A component; (*l*) D3 component); Carboniferous rocks: (m) A component; (n) C component (exposure 8); (o) C component (exposure 10); (p), (q) P component. The stereonet projections of the distribution of the A components are presented in geographical coordinates, for other components, in stratigraphic coordinates; open (filled) circles show the projections of the vectors on the upper (lower) hemisphere.

As seen in the Zijderveld diagrams (Fig. 2, samples 26-7, 29-6, 34-2, and 35-3, purple red siltstones and sandstones), the natural remanent magnetization of the studied samples has two components. Component A isolated in the temperature interval from 100 to 400°C is most likely to be associated with the viscous magnetization of the rocks in the Cenozoic geomagnetic field. The fold test for the A component is indeterminate. The unfolding test gives the direction of this component close to the present geomagnetic field at the sampling localities (D = 5°, I = 83°). The distribution of the directions of A is shown in Fig. 3a, and their statistical characteristics are presented in Table 1.

The high-temperature bipolar component D1 is isolated along the interval of 500–680°C. Figure 3b shows the distribution of the directions of this component, and their statistical characteristics are presented in Table 1. The components of both polarities are pre-folding and their fold tests are positive. The reversals test for D1 is positive and corresponds to class C (the angle between the axes is 10.8° with a critical angle of 11.4°).

Early Devonian Rocks. D_1p, Lower Part of Kapp Kjeldsen Formation.

The natural remanent magnetization of the studied samples varies within 1 to 15 mA/m about the average value of 5 mA/m. The magnetic susceptibility ranges from 192×10^{-6} to 425×10^{-6} SI units with the average of 294×10^{-6} SI units.

The Zijderveld diagrams (Fig. 2, samples 74, 89, 103, and 109, purple red siltstones) show that natural remanent magnetization of these samples has two components. Component A is isolated in the temperature interval from 100 to 450°C. Its increased thermal stability means that this component is not only associated with the viscous magnetization of the rocks in the Cenozoic geomagnetic field but is also due to the processes of weathering. The fold test for component A is negative. The distribution of the directions of this component is shown in Fig. 3c, and the statistical characteristics are presented in Table 1.

Table 1. Paleomagnetic directions for Devonian and Carboniferous sedimentary rocks of Spitsbergen Island

No.	ChRM age, Ma	Comp. J_n (intervals of isolation °C)	co-ord. sys.	*N*/*n*	D°	I°	*K*	α_{95}°	Test	Φ,°N	Λ,°E	dp°	dm°	φ_m°
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
loc. 1, D_1l, Liefdefjorden, (φ = 79.63° N, λ = 12.76° E)														
1	KZ	A (100–400°C)	g	41/65	34	73	24	5	F~(d/c = 0.4 ± 0.2)	67	146	8	9	59
		unfolding test	g		312	81	29	4	40%	77	276	8	8	72
2		D1 – (500–680°C)	g	17/18	199	47	9	12						
			s		211	5	16	9	F+(d/c = 1.1 ± 0.5)					
3		D1 + (500–680°C)	g	22/37	22	–29	12	9						
			s		27	5	20	7	F+(d/c = 1.0 ± 0.4)					
4	411–416	D1 (2 + 3)	g	39/55	21	–37	10	8	F+, R+, class C					
			s		29	0	17	6	(γ/γ_{cr} = 10.8°/11.4°)	9	163	3	6	0
loc. 2, D_1p, Bockfjorden, (φ = 79.45° N, λ = 13.46° E)														
5		A (100–450°C)	g	22/22	12	78	68	4	F–(d/c = 0.3 ± 0.4)	77	172	7	8	67
6		D2 – (550–680°C)	g	13/13	220	23	16	11						
			s		216	26	18	10						
7		D2+ (645–680°C)	g	4/4	26	–19	51	13						
			s		24	–19	60	12	R+, class C,					
8	407–411	D2 (6 + 7)	g	17/17	216	22	18	9	((γ/γ_{cr} = 13°/14°)					
			s		213	25	20	8	F~(d/c = 1.7 ± 1.4)	–4	161	5	9	–13
loc. 3, D_1p , fold, Bockfjorden, (φ = 79.52° N, λ = 13.4° E)														
9		A (100–550°C)	g	10/15	16	80	77	6	F–(d/c = 0.1 ± 0.2)	80	161	11	12	71
10		D2 – (640–680°C)	g	9/13	222	16	16	13						
			s		213	24	35	10						
11		D2 + (640–680°C)	g	1/2	27	–21								
			s		34	25								
12	407–411	D2 (10 + 11)	g	10/15	220	17	17	12	F+					
			s		213	24	29	9	(d/c = 1.0 ± 0.8)	–4	161	5	9	–13

Table 1. (Contd.)

No.	ChRM age, Ma	Comp. J_n (intervals of isolation °C)	co-ord. sys.	*N*/*n*	D°	I°	*K*	α_{95}°	Test	Φ,°N	Λ,°E	dp°	dm°	φ_m°
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D_1p, Bockfjorden, (φ = 79.45° N, λ = 13.46° E), (Iosifidi et al., 2007)														
13	KZ	A (100–550°C)	g	74/129	0	80	46	3	F–	81	194	6	7	71
14	407–411	D2–(640–678°C)	g	49/74	210	21	10	7	F~, R+, class C					
			s		208	22	10	7	(γ/γ_{cr} = 11°/13°)	–2	164	4	7	–11
D_1p, Bockfjorden														
15	KZ	A (loc. 1, 2, 3)	g	106/166	4	80	51	2	F–(d/c = 0.1 ± 0.2)	81	185	4	4	71
16	407–411	D2 (loc. 1, 2, 3)	g	76/106	33	21	12	5	F+(d/c = 0.7 ± 0.7)					
			s		30	–23	12	5	R+, class C	–3	164	3	5	–12
D_2g , Billefjorden (φ = 78.6° N, λ = 16.2° E)														
17	KZ	A (100–250°C)	g	32/32	359	85	8	10	F–(d/c = 0.1 ± 0.1)	82	199	18	19	71
18	392–382	D2–(300–530°C)	g	8/8	199	29	12	17						–5
			s		221	10	17	14	F+(d/c = 0.6 ± 0.4)	4	155	7	14	
19		P (300–530°C)	g	8/8	242	–4	5	28						
			s		216	–54	25	11	F+(d/c = 1.1 ± 0.4)					
20*			g	10/10	37	37	d = 43		F~					
			s		33	68	d = 18							
21	270–300	P (19 + 20)	g	18/18	228	–15	4	19						
			s		212	–58	28	7		48	158	8	10	39
D_3fr, Billefjorden (φ = 78.6° N, λ = 16.2° E)														
22	KZ	A (100–400°C, 100–600°C)	g	44/44	26	80	14	6	F–(d/c = 0.16 ± 0.21)	80	142	11	12	71
23		D3–(300–650°C)	g	21/21	210	–18	7	13						
			s		211	–31	32	6	F+(d/c = 1.0 ± 0.1)	26	163	4	7	17
24		D3+(400 –650°C)	g	8/8	38	35	39	9						
			s		39	35	45	8	F~					
25	370–390	D3 (23 + 24)	g	29/29	213	–23	9	10	F+, R+, class B					
			s		213	–32	33	5	(γ/γ_{cr} = 8°/10°)	27	161	3	6	17

Table 1. (Contd.)

No.	ChRM age, Ma	Comp. J_n (intervals of isolation °C)	co-ord. sys.	*N*/*n*	D°	I°	*K*	α_{95}°	Test	Φ,°N	Λ,°E	dp°	dm°	φ_m°
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$C_{1\text{-}2}$, Billefjorden, ($\varphi = 78.6°$ N, $\lambda = 16.2°$ E)														
26	KZ	A (100–450°C)	g	27/72	327	79	17	7	F~ (d/c = 04 ± 0.4)	77	256	13	13	69
		unfolding test	g		358	82	18	7	37%	86	203	13	14	74
27		C+, loc. 8 (500–700°C)	g	4/8	33	27	29	17						
			s		34	24	44	14						
28		C–, loc. 8 (500–680°C)	g	3/4	208	–27	33	22						
			s		208	–27	33	22						
29		C, loc. 8	g	7/12	31	27	36	10						
			s		32	25	44	9						
30		C+, loc. 10 (400–680°C)	g	14/32	11	29	13	11						
			s		23	27	16	10						
31		C–, loc.10 (400–685°C)	g	3/6	200	–37	30	23						
			s		211	–33	30	22						
32		C, loc. 10	g	17/38	12	30	14	10						
			s		24	28	18	9						
33	312–330	C (4 + 7)	g	24/50	18	29	15	8	F+, R+, class C					
			s		27	27	22	7	($\gamma/\gamma_{cr} = 5°/13°$)	24	167	4	8	14
34		P (500–680°C)	g	7/7	194	–64	56	8						
			s		201	–61	36	10						
35*		P (300–650°C)	g	11/11	41	63	d = 15°							
			s		41	63	d = 15°							
36	270–300	P (9 + 10)	g	18/18	185	–65	52	5						
			s		209	–61	40	6	F~	52	161	7	9	42

Note: *N* is the number of the hand samples; *n* is the number of the specimens; coord. sys. is the coordinate system: *g* geographical coordinates, *s* stratigraphical coordinates; *D*, *I* are the declination and inclination, respectively of the average direction of the components; *K* is concentration of vectors; α95° is the confidence circle radius with a probability of 95% for the average direction; * the direction is determined from the demagnetization circles; *F*+(–, ~) the fold test is positive (negative, uncertain); *R* polarity reversals test is +(–) positive (negative); Φ°, Λ° are the latitude and longitude of the paleomagnetic pole; dp°, dm° are the semiaxes of the confidence ellipsis; φ_m° is the paleomagnetic latitude; the statistic is estimated at a level of hand samples.

The high-temperature bipolar component D2 is isolated in the temperature intervals 550–678°C and 645–678°C. The reversals test for D2 component is positive and correspond to class C (the angle between the axes is 13° with a critical angle of 14°). The distribution of the directions of this component is illustrated in Fig. 3d, and their statistical parameters are presented in Table 1.

The samples acquired from the syncline fold (Fig. 1, outcrop 3) also have two magnetization components, whose directions are close to those determined in the samples from the exposures on the eastern coast of the Bockfjorden (Fig. 1, locality 2). The examples of the Zijderveld diagrams are shown in Fig. 2 (samples 147 and 159B). The distribution of the components isolated is displayed in Figs. 3e and 3f, and their statistical characteristics are indicated in Table 1. According to the fold test, the A component is post-folding, while component D2 is pre-folding. Table 1 also presents the paleomagnetic data from (Iosifidi et al., 2007) for the previously studied part of the Kapp Kjeldsen Formation at the sampling locality 2 (Fig. 1); rows 15 and 16 in Table 1 present the values averaged over all the data available for this formation.

Middle Devonian Rocks. D_2g, Estheriahaugen Formation

The natural remanent magnetization of the studied samples widely varies from 0.1 to 8.7 mA/m. The magnetic susceptibility of the rocks ranges from -5×10^{-6} to 270×10^{-5} SI units. The maximal values of J_n and K_m are found in the samples containing pyrite concretions. Under thermal demagnetization, all the samples underwent chemical alteration when heated to 350°C. This was mainly due to pyrite oxidation and the decomposition of coaly inclusions, which blocked the isolation of the characteristic component J_n in these samples. In some cases, it was possible to come to the emergence of the last component J_n (Fig. 2, sample 230). In this part of the collection, the second component (D2) is isolated in the interval from 300 to 580°C. In the other part of the samples, the second component (P) was identified as the average of the points in the interval 300–530°C (Fig. 2, sample 382), since at 550–580°C, the residual remanent vector behaves chaotically because of the additional magnetization from the newly formed magnetite, which is fixed by the increase in the magnetic susceptibility of the rocks. The P component is also clearly recognized by the demagnetization circles in the interval of 300–500°C. The distribution of the identified components is shown in Figs. 3h–3j, and their statistical characteristics are presented in Table 1. The fold test indicates that the A component is post-folding, whereas components D2 and P are pre-folding.

Late Devonian Rocks. D_3fr, Fiskeklofta Formation

The natural remanent magnetization of the studied samples widely varies from 0.2 to 60 mA/m. The magnetic susceptibility ranges from 6×10^{-6} to 340×10^{-6} SI units. The examples of the Zijederveld diagrams are shown in Fig. 2 (samples 295, 307, 327, 338, and 341). The characteristic component J_n was not possible to reveal from all the studied samples. At demagnetization, the heating to 300°C gives rise to chemical alterations, which is fixed by the growth of magnetic susceptibility. For example, after heating to 500°C, the magnetic susceptibility in samples 307 and 327 (Fig. 2) increases tenfold and fivefold, respectively, which makes it impossible to isolate the pure characteristic component J_n. A part of the samples was completely remagnetized by the Cenozoing geomagnetic field. Nevertheless, the bipolar component D3 isolated in some of these samples has a positive fold test and reversals test (corresponding to class B, the angle between the axes of normal and reversed polarities is 8° with a critical angle of 10°). The distribution of the identified components is illustrated in Figs. 3k and 3l. The statistical characteristics are presented in Table 1.

Carboniferous Rocks. Billefjorden

The natural remanent magnetization of the studied samples varies from 1 to 100 mA/m. The magnetic susceptibility ranges from 50×10^{-6} to 125×10^{-6} SI units. The examples of Zijderveld diagrams are presented in Fig. 2 (samples 398, 404, 405, 420, 421, and 427). The first component J_n is identified in the temperature interval from 100 to 450° and it is post-folding (Fig. 3m, Table 2). In the high-temperature interval, two components of J_n are recognized. The bipolar component C is isolated in the interval of 500–700°C, Fig. 2 (samples 404, 405, 421, and 427). Component C has positive reversals test (class C, the angle between the axes of normal and reverse polarities is 5° with a critical angle of 13°). Some samples are found to contain the component with reversed polarity, P, in the interval of temperature from 500–700°C (samples 398 and 420). The same component is also recognized by the demagnetization circles in the temperature intervals of 300–500°C and 600–655°C. According to the fold test, component A is post-folding, while components C and P are pre-folding. The distribution of the revealed components is shown in Fig. 3m–3q, and their statistical characteristics together with the coordinates of the paleomagnetic poles for the obtained J_n components are presented in Table 1.

In most of the studied Carboniferous and Devonian sections, normal and reverse polarity zones are revealed. The correlation of the studied sections to the general magnetostratigraphic scale is shown in Fig. 4. In order to assess the palaeogeographic relationship

Table 2. The paleomagnetic poles and APWP coordinates for the Russian Platform in Cambrian to Mesozoic

No.	Age, Ma	Coordinates of the pole		A_{95}°	Q	APWP		Data source
		Φ, °N	Λ, °E			Φ_s° N	Λ_s° E	
1	2	3	4	5	6	7	8	9
1	124.0	64	185	3	5	63	185	9228
2	138.0	67	168	7	3	68	172	4323
3	153.0	81	169	4	4	72	149	4354
4	160.0	67	134	3	6	69	134	Guzhikov et al., 2010
5	170.0	63	117	3	4	65	119	4357
6	200.0	70	88	7	4	57	100	4473
7	204.0	50	106	0	6	57	100	Yuan et al., 2011
8	213.5	38	98	4	5	47	103	6742
9	220.0	59	104	6	4	47	103	4643
10	229.0	56	149	4	3	54	134	4486
11	235.0	51	147	7	5	53	148	6746
12	242.0	53	158	4	5	50	159	4598
13	243.0	52	165	4	5	50	160	4546
14	248.0	44	162	6	5	47	162	4594
15	251.0	45	162	4	4	46	163	4601
16	251.5	43	160	5	4	46	163	4602
17	253.0	42	159	5	4	45	163	4599
18	253.5	45	168	6	4	45	163	4774
19	254.0	43	167	4	4	45	163	4753
20	254.0	46	169	5	4	45	163	4766
21	254.5	44	165	1	4	45	163	4770
22	254.5	45	157	3	4	45	163	4773
23	255.0	49	158	3	5	45	163	4767
24	255.0	44	166	3	5	45	163	4763
25	257.0	45	159	3	4	45	163	4779
26	258.0	45	157	3	4	45	163	4786
27	260.0	45	165	3	4	45	164	4782
28	261.0	45	163	6	4	45	164	4810
29	261.5	45	164	4	4	45	164	4809
30	261.5	45	169	6	4	45	165	4808
31	262.0	44	165	4	6	45	165	Shatsillo et al., 2006
32	266.0	47	163	2	4	45	166	4840
33	266.0	43	168	6	4	45	166	4693
34	267.0	47	166	4	4	45	166	4800
35	268.0	44	169	3	4	45	166	4799
36	269.0	43	167	2	4	45	166	4816
37	270.0	46	166	4	4	45	166	4811
38	280.0	44	160	3	4	44	165	7342
39	297.0	43	165	2	6	43	163	Iosifidi et al., 2010
40	310.5	44	150	10	4	41	157	4983
41	312.5	42	150	11	4	40	156	4961

Table 2. (Contd.)

No.	Age, Ma	Coordinates of the pole		A_{95}°	Q	APWP		Data source
		Φ, °N	Λ, °E			Φ_s° N	Λ_s° E	
1	2	3	4	5	6	7	8	9
42	315.0	35	162	3	5	39	156	Meijers et al., 2010
43	375.0	11	153	8	4	9	141	6759
44	376.0	11	134	3	5	8	141	Gurevich et al., 2005
45	377.0	5	143	3	4	7	141	6855
46	384.0	–1	145	8	5	0	140	6859
47	385.0	–3	136	4	5	–1	140	6854
48	405.5	–4	146	7	6	–5	138	6958
49	410.0	–7	123	10	5	–6	139	6866
50	425.0	–7	175	7	6	–7	171	Lubnina et al., 2007
51	460.0	–12	222	5	4	–13	226	Rodionov, Gurevich, and Pavlov, 2002
52	468.0	–9	239	6	5	–18	234	Lubnina et al., 2005*
53	472.0	–35	239	6	5	–22	235	8254
54	474.0	–25	238	3	5	–22	236	Gurevich et al., 2005
55	475.0	–24	228	7	4	–23	237	Gurevich et al., 2005
56	477.0	–18	235	6	6	–23	237	Khramov and Iosifidi, 2009
57	500.0	–22	267	5	5	–22	266	Khramov and Iosifidi, 2009

Note: Age (Ma) is indicated according to (*Stratigraficheskii...*, 2006); A_{95} is the confidence circle radius with a probability of 95% for the location of paleomagnetic pole; *Q* is the sum of the Van der Voo reliability criteria (Van der Voo, 1990); Φ°, Λ° are the latitude and longitude of the north paleomagnetic pole; Φ_s°, Λ_s° are the latitude and longitude of the north paleomagnetic pole for APW path determined by the spherical spline method (Jupp and Kent, 1987) with a smoothing parameter of 200 and normalization to *Q* factor. 9228, 8254... are the codes of the results (RESULTNO) from the GPMDP International database of paleomagnetic data (McElhinny and Lock, 1996; Pisarevsky, 2005); * is the average pole over four sections.

between Spitsbergen and the Russian Platform, we compared the paleomagnetic results for these tectonic units. The apparent polar wander paths (APWP) estimated by the spherical spline technique (Jupp and Kent, 1987) for the Russian Platform in the Cambrian–Cretaceous and for Spitsbergen in the Devonian–Cretaceous are presented in Tables 2 and 3, respectively. The mutual positions of the apparent polar wander paths (APWP) for the Russian Platform and Spitsbergen in the interval from Devonian to Triassic are displayed in Fig. 5. The data for Spitsbergen were converted to the coordinates of the Russian Platform by matching the curves for the coeval paleomagnetic poles (two coeval poles from each APWP) on two time intervals: 240–265 Ma and 345–395 Ma. The following poles of rotation are identified: I (Permian–Triassic APWP segment) with $\Phi = 71^{\circ}$ N, $\Lambda = 335^{\circ}$ E, and angle of rotation -14°; II (Devonian–Carboniferous APWP segment) with $\Phi = 71^{\circ}$ N, $\Lambda = 163^{\circ}$ E, and angle of rotation -19°. The closeness (the degree of agreement) between the coeval paleomagnetic poles is estimated by the angular distance between them along the great circle arc calculated by the GCD method (Torsvik et al., 2001). In both cases, this deviation was (9 ± 4) degrees. The layout of EEP (according to the data for the Russian Platform) and Svalbard Archipelago for the Early Devonian (~405 Ma) and Early Triassic (~240 Ma), reconstructed from the paleomagnetic data (Tables 2 and 3), is shown in Fig. 6. For the Early Devonian and Early Triassic, we used the poles of rota-

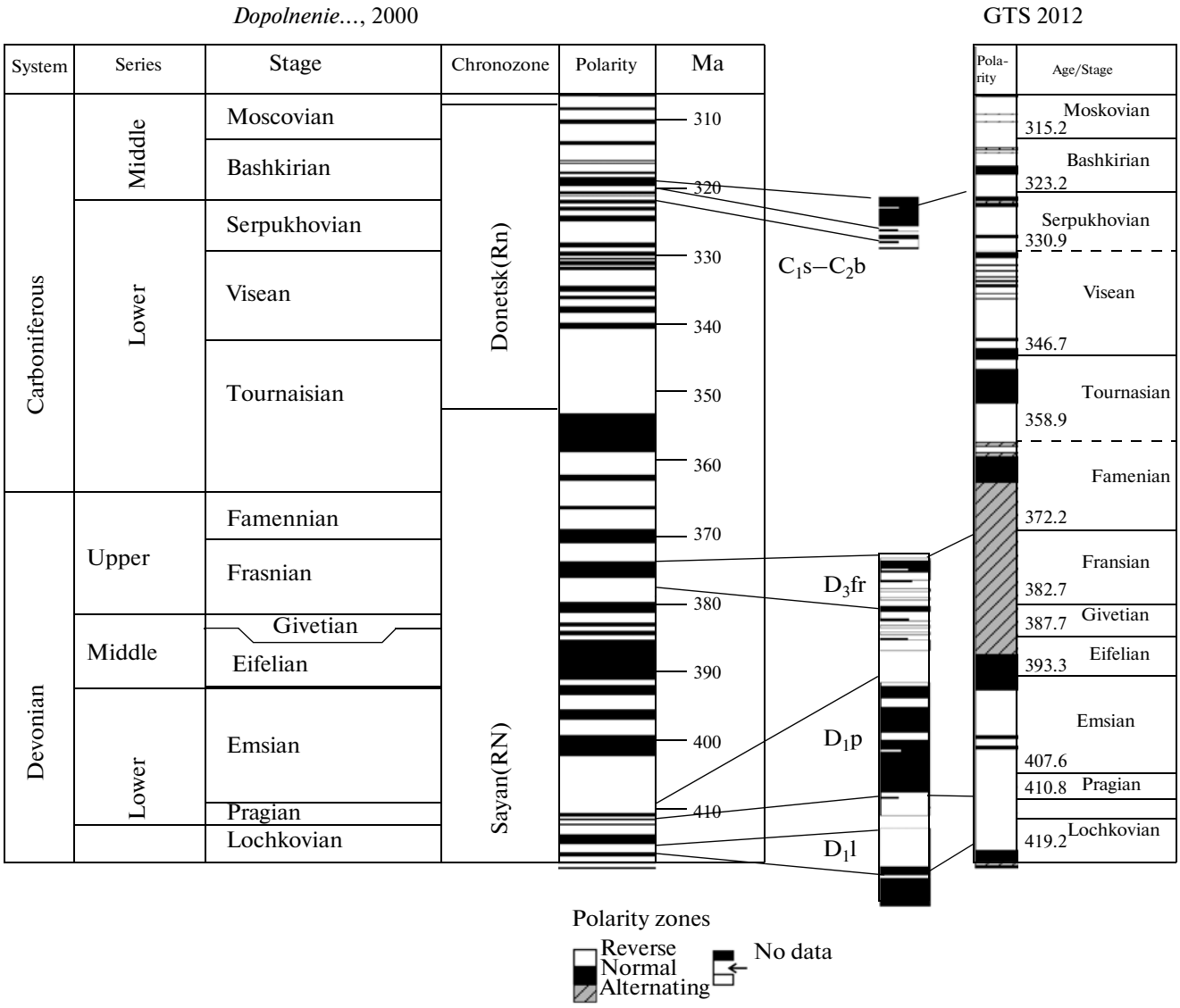


Fig. 4. The correlation of the identified magnetozones to the general magnetostratigraphic time scale.

tion for variants II and I, respectively. The current pattern of the positional relationship of these tectonic units lies somewhere between the first and second variants. The relative position of EEP (according to the data for the Russian Platform) and Svalbard for the Early Triassic agrees with the second model of palaeogeographic relationship between these blocks suggested in (Torsvik et al., 2001), and their relative positions in the Early Devonian agree with the first model.

DISCUSSION

The component analysis of the natural remanent magnetization of the Devonian and Carboniferous sedimentary rocks of Spitsbergen revealed two-component J_n in all samples. Component A, isolated in a wide temperature interval (100–350°C, 100–450°C, and 100–550°C), is most likely to be related to both the viscous magnetization of the rocks by the Cenozoic geomagnetic field and the chemical alterations caused by weathering. The second high-temperature component in most of the sections is bipolar and pre-folding. The zones of normal and reverse polarity identified in our study for the Fiskeklofta Formation are coherent with the results obtained by other authors (Kolesov, 1985; Pogarskaya and Gurevich, 1988). The positive reversals tests and the reasonable correspondence between the locations of the zones of normal and reverse polarity revealed in our study and the general magnetostratigraphic scale (Fig. 4) also support the old age of the isolated characteristic components of J_n. It should be noted that isolating the characteristic components of J_n

Table 3. The paleomagnetic poles and APWP coordinates for Spitsbergen Island in Devonian to Mesozoic

No.	Age, Ma	Coordinates of the pole		A_{95}°	Q	APWP				Data source
						Φ_s° N	Λ_s° E	in the coordinates of the Russian Platform		
		Φ, °N	Λ, °E					Φ_r° N	Λ_r° E	
1	110.0	70	223	8	5					74
2	120.0	66	200	7	6					6220
3	145.0	53	154	10	4	53	154	52	135	Gurevich and Slautsitais, 1988*
4	223.4	49	160	6	4	50	166	50	148	Gurevich and Slautsitais, 1988**
5	241.0							50	159	
6	250.0	48	182	11	4	48	181	49	163	8300
7	290.0	38	171	8	5	38	171	38	155	8301
8	325.0	24	167	6	5	25	167	25	152	this work
9	343.0	27	164	11	4	27	165	28	150	1132***
10	374.0	27	161	5	5	24	160	25	145	this work
11	385.0	4	155	10	4	10	157	12	140	this work
12	403.0							−2	142	
13	410.0	−3	164	4	6	−1	164	0	146	this work
14	414.5	−2	172.3	3	4	2	166	3	148	7368
15	416.0	9	163	6	5	4	166	4	149	this work

Note: Φ_r°,, Λ_r° are the latitude and longitude of the north paleomagnetic pole for APW path in the coordinates of the Russian Plate. The rotation about the rotation pole $\Phi_e = 7°$ N, $\Lambda_e = 335°$ E by 14° clockwise for the poles starting from 325 Ma and younger; rotation about the rotation pole $\Phi_e = 71°$ N, $\Lambda_e = 363°$ E by 19° clockwise for the poles starting from 343 Ma and older; * is the average pole over five sections; ** is the average pole over two sections; *** is the direction for *R* polarity (Watts, 1985); remaining notations are the same as indicated in the Note to Table 2.

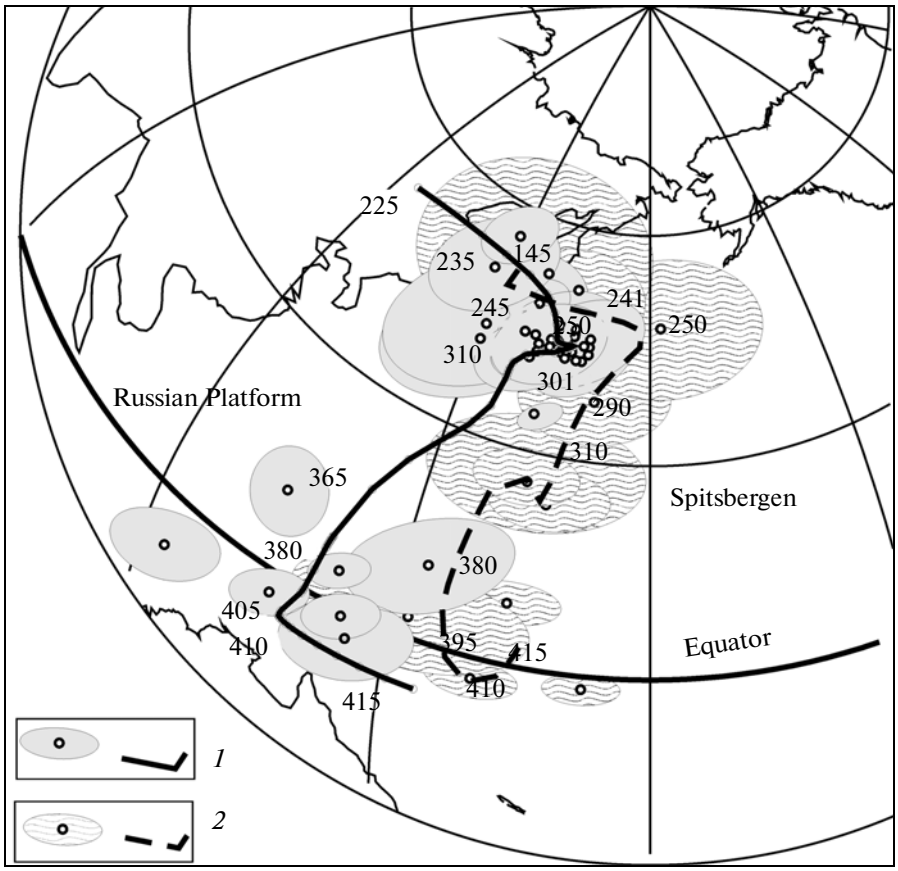


Fig. 5. The relative position of APWP for the Russian Platform and Spitsbergen in Devonian–Triassic: *1* the data from the Russian Platform; *2* the data from Spitsbergen; the initial data are shown by the confidence ellipses of the positions of paleomagnetic poles.

in the Middle and Upper Devonian rocks is troublesome due to chemical alterations occurring during the thermal demagnetization. The difficulty in isolation of characteristic component of J_n in the Upper Devonian series of Spitsbergen was also noted by other authors (Kolesov, 1985; Pogarskaya and Gurevich, 1988). The probable remagnetization of red Devonian rocks was pointed out in (Lovlie et al., 1984; Torsvik, Lovlie, and Sturt, 1985). In the northern sections, which are dominated by redbeds, the statistically significant components associated with the Late Paleozoic remagnetization are not revealed. The Late Paleozoic remagnetization is identified in the Middle Devonian and Carboniferous rocks from the sections located further south (P component).

The paleomagnetic pole from the Lower Devonian rocks (the Kapp Kjeldsen Formation) closely agre es with that determined by Pogarskaya and Gurevich (1988), who applied the method of common intersection of the remagnetization planes for the data from three exposures ($\Phi = 2°$ S, $\Lambda = 172°$ E). Our analysis managed to isolate the characteristic component D1 from the results of thermal demagnetization. The paleomagnetic pole determined from the D1 component falls in the Early Devonian segment of APWP for EEP and lies close to the key paleomagnetic pole for the Early Devonian (Smethurst and Khramov, 1992) $\Phi = 4°$ S, $\Lambda = 146°$ E, dp = 5, dm = 9. The reconstructed trajectories of the apparent wandering of the paleomagnetic pole for the Russian Platform and Spitsbergen have similar shapes (Fig. 5).

The relative position of EEP (according to the data for the Russian Platform) and Svalbard Archipelago, which is shown in Fig. 6, is consistent with the geological evidence for the evolution of this region. In the Devonian, Laurussia continued to form and converge with Gondwana, which has resulted in the formation of Pangaea at the Visean Stage of Carboniferous (Ziegler, 1989). At the final stage of this process, the left-lateral shift between Laurentia and EEP occurred, which was also reflected in the corresponding shear deformations within Spitsbergen. This process was accomplished by

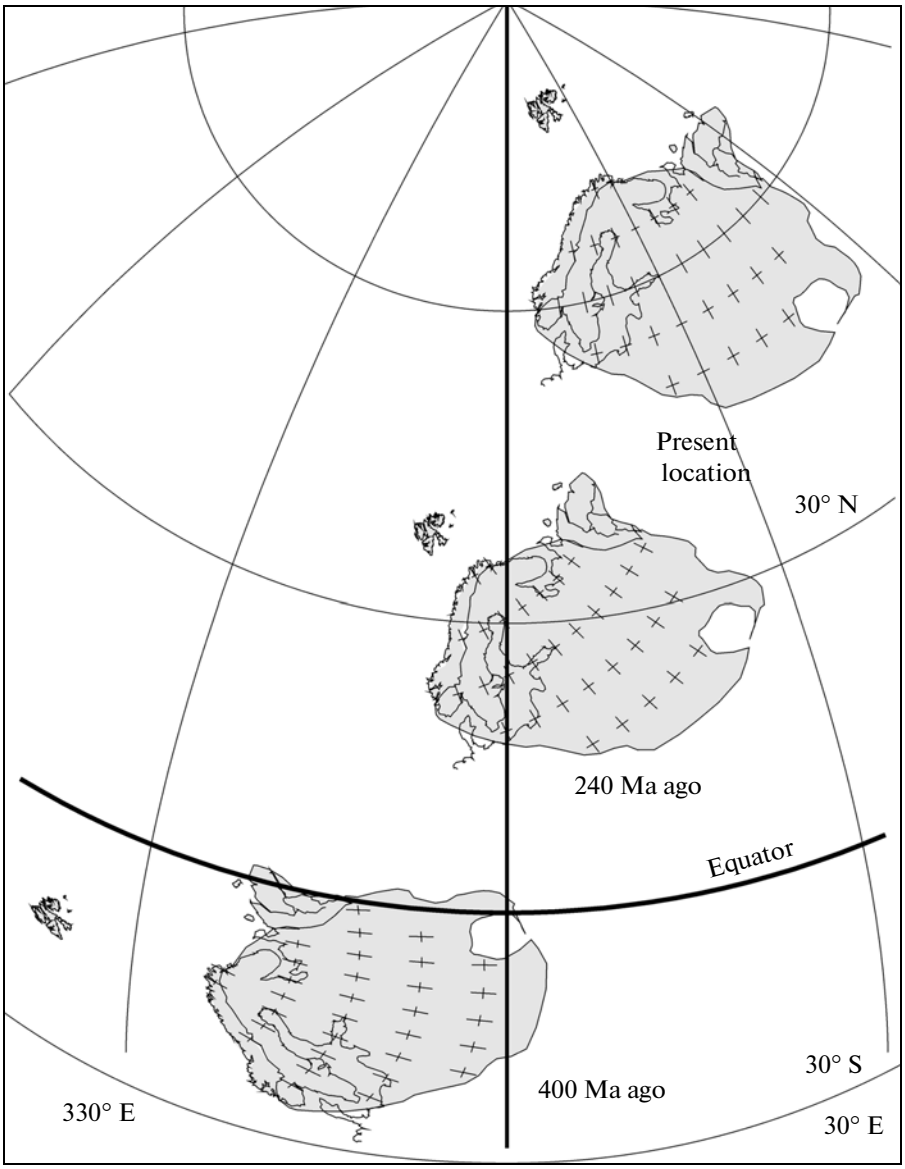


Fig. 6. The relative position of the Russian Platform and Spitsbergen in Early Devonian and Early Triassic. The following paleomagnetic poles of the Russian Platform and Spitsbergen are used for the reconstruction (Tables 2 and 3): Early Triassic $\Phi = 50°$ N, $\Lambda = 159°$ E; Early Devonian $\Phi = 5°$ S, $\Lambda = 138°$ E (the Russian Plate) and $\Phi = 2°$ S, $\Lambda = 142°$ E (Spitsbergen) taken from the APWP splines.

the Late Permian and, correspondingly, it caused the convergence between Spitsbergen and EEP (Fig. 6). In the course of the further northward drift and breakup of Pangaea, the relative location of EEP and Svalbard Archipelago has approached its present geometry (Fig. 6).

CONCLUSIONS

Our studies of the Devonian and Carboniferous sedimentary rocks of Spitsbergen demonstrate the possibility to isolate the old components of natural remanent magnetization in the rocks on this island. This is confirmed by the positive field tests and reasonable agreement between the positions of normal and reverse polarity zones identified in four sections and the general magnetostratigraphic scale. In contrast to the Barents-Pechora region (Pai Khoi and Polar Urals), the Late Paleozoic remagnetization of the older deposits is less pronounced in our case.

The reconstructed paleogeographic position of Spitsbergen and EEP in the Early Devonian and Early Triassic demonstrates the common north-northeastward drift of these units from the southern equatorial latitudes in Early Devonian to the middle northern latitudes with a small, compared to this drift, convergence

of Spitsbergen and EEP (800 km) by the Early Triassic and its subsequent retrograde motion.

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